MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11746960 SWITCH AND LIGHT CONTROL

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly: 11746960 Switch and Light Control which is the A10 component of the Electronic Unit, 11743131.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment: Acquisition, Maintenance and Disposition of
MIL-STD-45662	Calibration System Requirements

FSC: 1240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

11746960	Circuit Card Assembly, Switch and Light Control

Inspection Equipment

11750202	Automatic Card and Unit, Test Set
11750437	Adapter, Switch and Light Control
11821457	Fixture Shock, Vertical Horizontal Plane
11821458	Shock Adapter, Plate Vib Horz-Vert Plane/ Shock-Z
11821520	Automatic Test Assembly Central Processor Unit
11821647	Vibration/Shock Test Fixture - Electronic Unit, AC, FAT
11821861	Test Fixture - Auto Card Test Set, Electronic Unit, FAT

Packaging Data Sheet

P11746960	Packaging of Circuit Card Assembly: 11746960 Switch and Light Control

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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shall be performed in an order satisfactory to the Government.

3. REQUIREMENTS

3.1 Fabrication. The switch and light control circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 11746960 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Function. The assembly shall provide the following conditions:

	Designation	Nomenclature
a.	0.4 Vdc	Lamp test
b.	0.4, 14.0 Vdc	Select
c.	0.2, 4.2 Vdc	Overrange
d.	4.2, 0.2 Vdc	Ripple blanking input
e.	4.2, 0.2 Vdc	Feed
f.	0.2, 0.4, 4.2, 14.0 Vdc	ON
g.	0.4, 14.5, 14.0, 0.4 Vdc	AUTO
h.	0.4, 14.5, 0.2 and 14, 0.4, 4.2 Vdc	TEST switch
i.	4.2, 0.2 Vdc	Range select
j.	0.2, 0.4, 4.2, 14.0 Vdc	Manual reset and range light signal
k.	4.2, 0.4, 0.2 Vdc	Turn-on reset
l.	4.2, 0.2, Vdc	RANGE
m.	14.0, 0.4, 4.2, 0.2 Vdc	GO, Battle range
n.	2.0 ohms	Continuity

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	73° ± 18°F
b.	Relative humidity	50 percent ± 30 percent
c.	Atmospheric pressure	28.5 + 2.0 -3.0 in. Hg.

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3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

3.3.1 Lamp test. With a logical zero digital type A signal, item 3.1, table I applied between pins P1-A30 and P1-A16 (rtn), the voltage between the following pins and P1-A16 (rtn) shall be 0.4 ±0.4 Vdc. (See 4.6.2.1)

a. P1-B7
b. P1-B10
c. P1-B11
d. P1-B20
e. P1-B22
f. P1-B23
g. P1-B25
h. P1-B29
i. P1-B30
j. P1-B32

TABLE I. Loads, power, and signals.

Item	Conditions	Characteristics	Connections
1.0	Loads		Connected as follows:
1.1	Resistor	2.4 kilohms ±5 percent, 1/2 Watt (W)	Between pins P1-A32 and P1-A16, as specified
1.2	Resistors	240 ohms ±5 percent, 2W, in series with item 2.2	One to each of pins P1-B20, P1-B22, P1-B23, P1-B25, and P1-B30
1.3	Resistors	180 ohms ±5 percent, 2W, in series with item 2.2	One to each of pins P1-B7, P1-B10, P1-B11, P1-B29, and P1-B32
1.4	Resistor	3.0 kilohms ±5 percent, 1/4 W	Applied between P1-A15 (+5 Vdc) and each of pins P1-A18, P1-A22, P1-A23, P1-A8, P1-A24 and P1-A35
1.5	Resistor	2.2 kilohms ±5 percent, 1/4 W	Applied between pin P1-A15 (+5 Vdc) and pin P1-A11
1.6	Resistor	1.0 kilohms ±5 percent, 1/4 W	Applied between pin P1-A15 (+5 Vdc) and pin P1-A33

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TABLE I. Loads, power, and signals. - Continued

Item	Conditions	Characteristics	Connections
1.7	Resistor	1.1 kilohms ±5 percent, 1/4 W	Applied between pin P1-A15 (+5 Vdc) and each of pins P1-A4, P1-B5, P1-B4, P1-B6, P1-A7, P1-A31, P1-A6 and P1-B26
1.8	Resistor	560 ohms ±5 percent, 1/4 W	Applied between pin P1-A15 (+5 Vdc) and pin P1-A29
2.0	Power		
2.1	15.0 Vdc	Tolerance: ±0.7 Vdc Ripple: 25 millivolts (mv) peak-to-peak (p-p) max	Applied between pins P1-A14 (+) and P1-A16 (rtn), as specified
2.2	14.0 Vdc	Tolerance: ±0.5 Vdc Ripple: 25 mV p-p max	Applied between pin P1-A16 (rtn) and each of items 1.2 and 1.3 resistors
2.3	5.0 Vdc	Tolerance: ±0.2 Vdc Ripple: 25 mV p-p max	Applied between pin P1-A16 (rtn) and pin P1-B15
3.0	Signals		
3.1	Digital type-A	Logical one: 4.0 ±1.0 Vdc Source Impedance: 2.0 kilohms ±5%	Applied as specified
		Logical zero: 0.2 ±0.2 Vdc. When applicable, pulse width and pulse-repetition frequency (PRF), as specified	Applied as specified
3.2	Digital type-B	Logical one: 5.0 ±1.0 Vdc Source Impedance: 51 ohms ±5%	Applied as specified
		Logical zero: 0.2 ±0.2 Vdc. When applicable, pulse width and PRF, as specified	Applied as specified

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3.3.2 Select. With a logical zero digital type-A signal, item 3.1 of table I, applied between pins P1-A28 and P1-A16 (rtn) the output voltage between pins P1-B30 and P1-A16 (rtn) shall be 0.4 ±0.4 Vdc. With the logical zero signal removed the voltage at pin P1-B30 shall change to 14.0 ±0.5 Vdc. (See 4.6.2.2)

3.3.3 Overrange. With the table II items input conditions established in any item order, the corresponding overrange output voltages shall be as specified therein. (See 4.6.2.3)

3.3.4 Ripple blanking input (RBI). With the table III items input conditions established in any item order, the corresponding $\overline{RBI}$ output voltages shall be as specified therein. (See 4.6.2.4)

TABLE II. Overrange outputs.

Item	Input conditions at pins 1/, 2/				Output voltage at pin P1-A23 1/ (Vdc)
	P1-A1	P1-A2	P1-A3	P1-A4	
1	1	1	1	0	0.2 ±0.2
2	1	1	0	1	0.2 ±0.2
3	1	1	0	0	0.2 ±0.2
4	1	0	1	1	0.2 ±0.2
5	1	0	1	0	4.2 ±1.0
6	1	0	0	1	4.2 ±1.0
7	1	0	0	0	4.2 ±1.0
8	0	1	1	1	4.2 ±1.0

1/ Input signal and output voltages referenced to pin P1-A16.
2/ Input signal condition legend
1 = logical one digital type-A signal, item 3.1 of table I.
0 = logical zero digital type-A signal, item 3.1 of table I.

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TABLE III. Ripple blanking input responses.

Item	Item conditions at pins 1/, 2/			Output voltage at pin P1-A22 1/ (Vdc)
	P1-A19	P1-A20	P1-A28	
1	1	1	1	4.2 ±1.0
2	1	1	0	0.2 ±0.2
3	1	0	1	0.2 ±0.2
4	1	0	0	0.2 ±0.2
5	0	0	0	4.2 ±1.0

1/ Input signal and output voltages referenced to pin P1-A16.
2/ Input signal condition legend
1 = logical one digital type-A signal, item 3.1 of table I.
0 = logical zero digital type-A signal, item 3.1 of table I.

3.3.5 $\overline{\text{Feed}}$. With a logical zero digital type-A signal, item 3.1 of table I, applied between pins P1-B3 and P1-A16 (rtn) the output voltage between pins P1-A8 and P1-A16 (rtn) shall be 4.2 ±1.0 Vdc. With the logical zero signal removed, the voltage at pin P1-A8 shall change to 0.2 ±0.2 Vdc. (See 4.6.2.5)

3.3.6 $\overline{\text{ON}}$. With a logical zero digital type-A signal, item 3.1 of table I, applied between pins P1-A12 and P1-A16 (rtn), the output voltage between pin P1-A11 and P1-A16 and pins P1-B20 and P1-A16 (rtn) shall be 0.2 ±0.2 Vdc and 0.4 ±0.4 Vdc, respectively. With the logical zero signal removed, the voltages at pins P1-A11 and P1-B20 shall change to 4.2 ±1.0 Vdc and 14.0 ±0.5 Vdc, respectively. (See 4.6.2.6)

3.3.7 $\overline{\text{AUTO}}$. With a logical zero digital type-A signal, item 3.1 of table I, applied between pins P1-B24 and P1-A16 (rtn) the outputs between pins P1-B25 and P1-A16 (rtn) and pins P1-A32 and P1-A16 (rtn) shall be 0.4 ±0.4 Vdc and 14.5 ±1.5 Vdc, respectively. With the logical zero signal removed, the voltages at pins P1-B25 and P1-A32 shall change to 14.0 ±0.5 Vdc and 0.4 ±0.4 Vdc, respectively. (See 4.6.2.7)

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3.3.8 TEST switch. With a logical zero digital type-A signal item 3.1 of table I, applied between pins P1-B19 and P1-A16 (rtn) the output voltages shall be as follows:

a. Between pins P1-B22 and P1-A16 (rtn); 0.4 +0.4 Vdc.
b. Between pins P1-A32 and P1-A16 (rtn); +14.5 +1.5 Vdc.
c. Between pins P1-A33 and P1-A16 (rtn); 0.2 ±0.2 Vdc.

With the change of input voltage between pins P1-B19 and P1-A16 (rtn) to a logical one, item 3.1, table I, the output voltages shall be as follows:

d. Between pins P1-B22 and P1-A16 (rtn); 14 ±0.5 Vdc.
e. Between pins P1-A32 and P1-A16 (rtn); 0.4 ±0.4 Vdc.
f. Between pins P1-A33 and P1-A16 (rtn); 4.2 ±1.0 Vdc. (See 4.6.2.8)

3.3.9 Range select. With the table IV input conditions established in the item number order, the corresponding ordered outputs shall be as specified therein. (See 4.6.2.9)

3.3.10 Manual reset and range light signal. With the table V input conditions established in any item order, the corresponding outputs shall be as specified therein. (See 4.6.2.10)

3.3.11 Turn-on reset. With pins P1-A14 and P1-A16 interconnected for not less than one second and then disconnected, subsequent application of the 15.0 Vdc power, item 2.1 of table I, between pins P1-A14 and P1-A16 (rtn) shall result in an appearance of the strobe output between pins P1-A18 and P1-A16 (rtn) and the manual reset reply gating output between pins P1-B26 and P1-A16 (rtn) as specified in figure 1 and the outputs at pins as specified in table VI. (See 4.6.2.11)

3.3.12 RANGE. With the table VII items input conditions established in any item number order, the outputs at pins as listed in table VII shall be as specified. (See 4.6.2.12)

3.3.13 GO, Battle range. With the table VIII items input conditions established in any item order, the outputs at pins as listed in table VIII shall be as specified. (See 4.6.2.13)

3.3.14 Continuity. With no power or signals applied, direct continuity, resistance not greater than 2.0 ohms, shall exist between pin P1-B14 and each of pins P1-A16, P1-B16, P1-A36 and P1-B36. (See 4.6.2.14)

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TABLE IV. Range select.

	Inputs conditions at pins 1/, 2/					
Item	P1-B9	P1-B12	P1-B8	P1-A20	P1-B2	P1-B1
1	1	1	1	1	0 3/	1
2	1	0	1	0	1	1
3	1	0	1	1	1	1
4	1	1	0	0	1	1
5	1	1	0	1	1	1
6	0	1	1	0	1	1
7	0	1	1	1	1	1
8	1	1	1	1	1	1 3/

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TABLE IV. Range select. - Continued

Item	Output voltages (Vdc) at pins 1/					
	P1-B11	P1-B5	P1-B10	P1-B6	P1-B7	P1-B4
1	14.0 ± 0.5	0.2 ± 0.2	14.0 ± 0.5	0.2 ± 0.2	0.45 ± 0.45	4.2 ± 1.0
2	14.0 ± 0.5	0.2 ± 0.2	14.0 ± 0.5	0.2 ± 0.2	0.45 ± 0.45	4.2 ± 1.0
3	14.0 ± 0.5	0.2 ± 0.2	0.45 ± 0.45	4.2 ± 1.0	14.0 ± 0.5	0.2 ± 0.2
4	14.0 ± 0.5	0.2 ± 0.2	0.45 ± 0.45	4.2 ± 1.0	14.0 ± 0.5	0.2 ± 0.2
5	14.0 ± 0.5	0.2 ± 0.2	14.0 ± 0.5	0.2 ± 0.2	0.45 ± 0 .45	4.2 ± 1.0
6	14.0 ± 0.5	0.2 ± 0.2	14.0 ± 0.5	0.2 ± 0.2	0.45 ± 0 .45	4.2 ± 1.0
7	0.45 ± 0.45	4.2 ± 1.0	14.0 ± 0.5	0.2 ± 0.2	14.0 ± 0.5	0.2 ± 0.2
8	14.0 ± 0.5	0.2 ± 0.2	14.0 ± 0.5	0.2 ± 0.2	0.45 ± 0.45	4.2 ± 1.0

1/ Input signals and output voltages referenced to pin P1-A16
2/ Input signal condition legend
1 = logical one digital type - A signal, item 3.1 of table I.
0 = logical zero digital type - A signal, item 3.1 of table I.
3/ This condition set after all other conditions established.

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TABLE V. Reset outputs.

Item	Input conditions at pins 1/, 2/		Output voltages at pins 1/ (Vdc)			
	P1-A25	P1-B1	P1-A29	P1-A7	P1-B26	P1-B29
1	0	0	0.2 ± 0.2	0.2 ± 0.2	0.2 ± 0.2	0.4 ± 0.4
2	1	1	4.2 ± 1.0	4.2 ± 1.0	4.2 ± 1.0	14.0 ± 0.5

1/ Input signal and output voltages referenced to pin P1-A16.
2/ Input signal condition legend
1 = Logical one digital type - A signal, item 3.1 of table I.
0 = Logical zero digital type - A signal, item 3.1 of table I.

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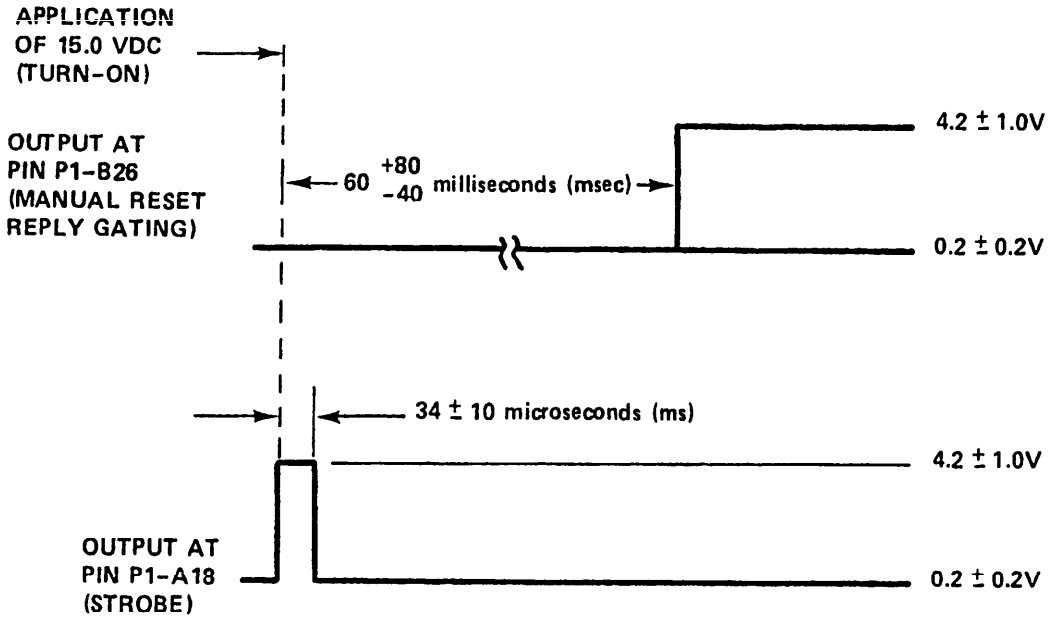


FIGURE 1. Turn-on reset.

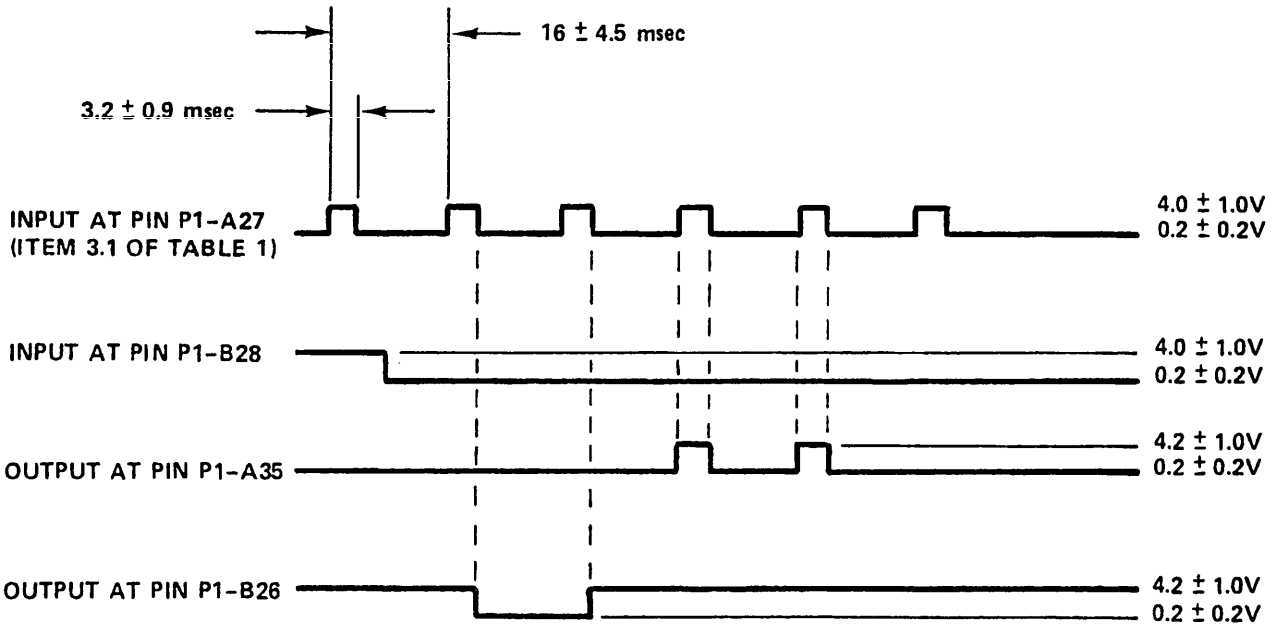


FIGURE 2. Range waveforms.

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TABLE VI. Turn-on reset.

Item	Pins of P1	Output Voltage 1/ Vdc
1	A31	4.2 ± 1.0
2	B32	0.4 ± 0.4
3	B33	0.2 ± 0.2

1/ Output voltages referenced to pin P1-A16 (rtn).

TABLE VII. Range switch.

	Input conditions at pins of P1 1/ 2/ 3/					
Item	B1	B2	B35	B34	A27	B28
1	0	NC	0	NC	NC	NC
2	0	NC	1	0*	NC	NC
3	0	NC	0	NC	NC	NC
4	1	Fig. 3	0	NC	NC	NC
5	NC	NC	0	NC	Fig. 2	Fig. 2

TABLE VII. Range switch. - Continued

	Outputs at pins of P1 1/ 2/ (Vdc)				
Item	A24	A35	A33	B26	B33
1	+4.2 ± 1.0	+0.2 ± 0.2	+4.2 ± 1.0	N/A	+0.2 ± 0.2
2	+0.2 ± 0.2	+4.2 ± 1.0	N/A	N/A	+0.2 ± 0.2
3	+4.2 ± 1.0	+0.2 ± 0.2	N/A	N/A	+0.2 ± 0.2
4	N/A	N/A	N/A	Fig. 3	Fig. 3
5	N/A	Fig. 2	+0.2 ± 0.2	Fig. 2	N/A

1/ Input signal and output voltages referenced to pin P1-A16.
2/ Input signal condition legend:
1 = Logical one digital Type A - Signal, Item 3.1 of table I.
0 = Logical zero digital Type A - Signal, Item 3.1 of table I.
0* = Logical zero digital Type A - Signal momentarily applied.
3/ NC = No Connection.
4/ N/A = Not Applicable.

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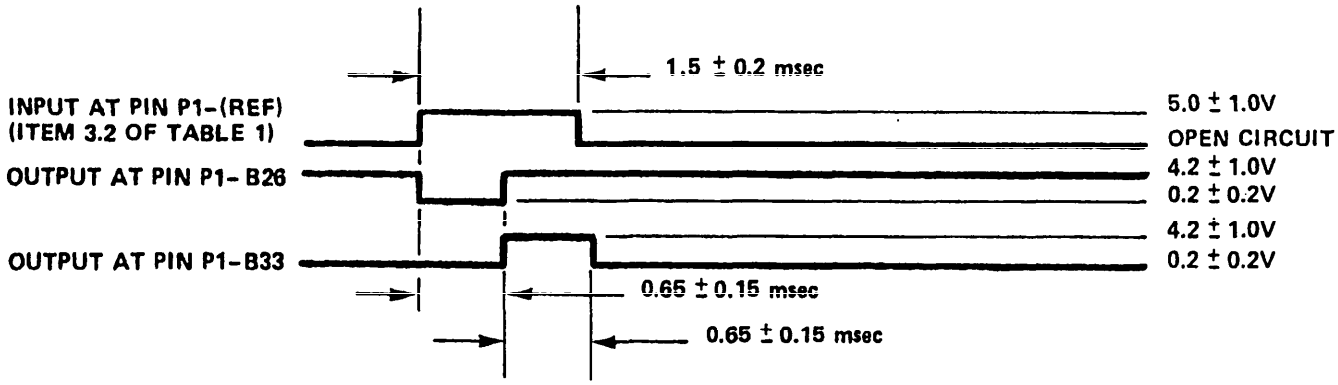


FIGURE 3. Range waveforms.

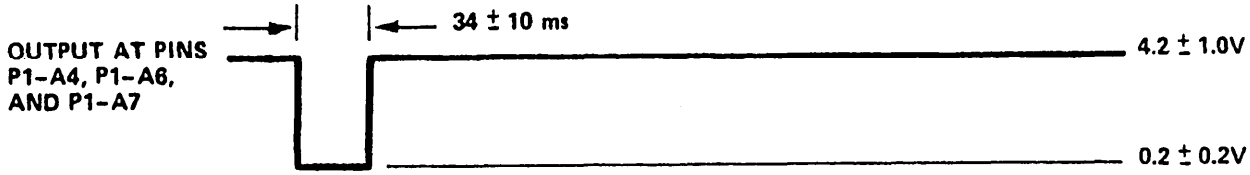


FIGURE 4. Manual reset select, 200 and 1000 signal waveform.

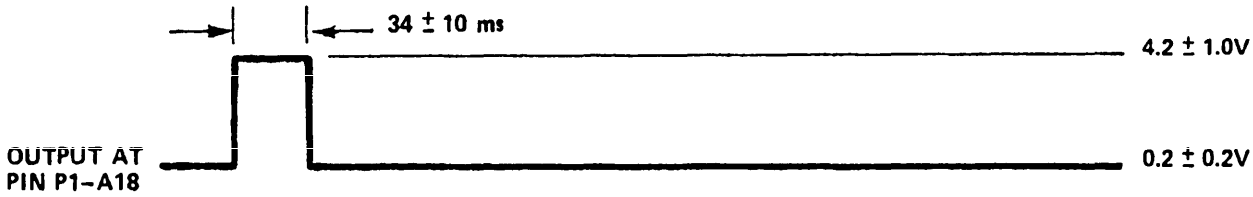


FIGURE 5. Strobe signal waveform.

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TABLE VIII. $\overline{GO}$, $\overline{BATTLE\ RANGE}$ requirements.

Item	Input conditions at pins of P1 1/, 2/		Outputs at pins of P1 1/ (Vdc)							
	A20	B27	B23	A31	B32	A4	A6	A18	A7	B26
1	1	0	14.0 ± 0.5	4.2 ± 1.0	0.4 ± 0.4	Fig. 4	Fig. 4	Fig. 5	Fig. 4	0.2 ± 0.2
2	0	1	0.4 ± 0.4	0.4 ± 0.4	14.0 ± 0.5	4.2 ± 1.0	4.2 ± 1.0	Fig. 5	4.2 ± 1.0	4.2 ± 1.0

1/ Input signal and output voltages referenced to pin P1-A16.
2/ Input signal condition legend:
1 = Logical one digital type - A signal, item 3.1 of table I.
0 = Logical zero digital type - A signal, item 3.1 of table I.

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3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity unit (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 120 ± 12 g for a duration of 1.7 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 6. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See .6.3.2)

3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +125°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to 160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5).

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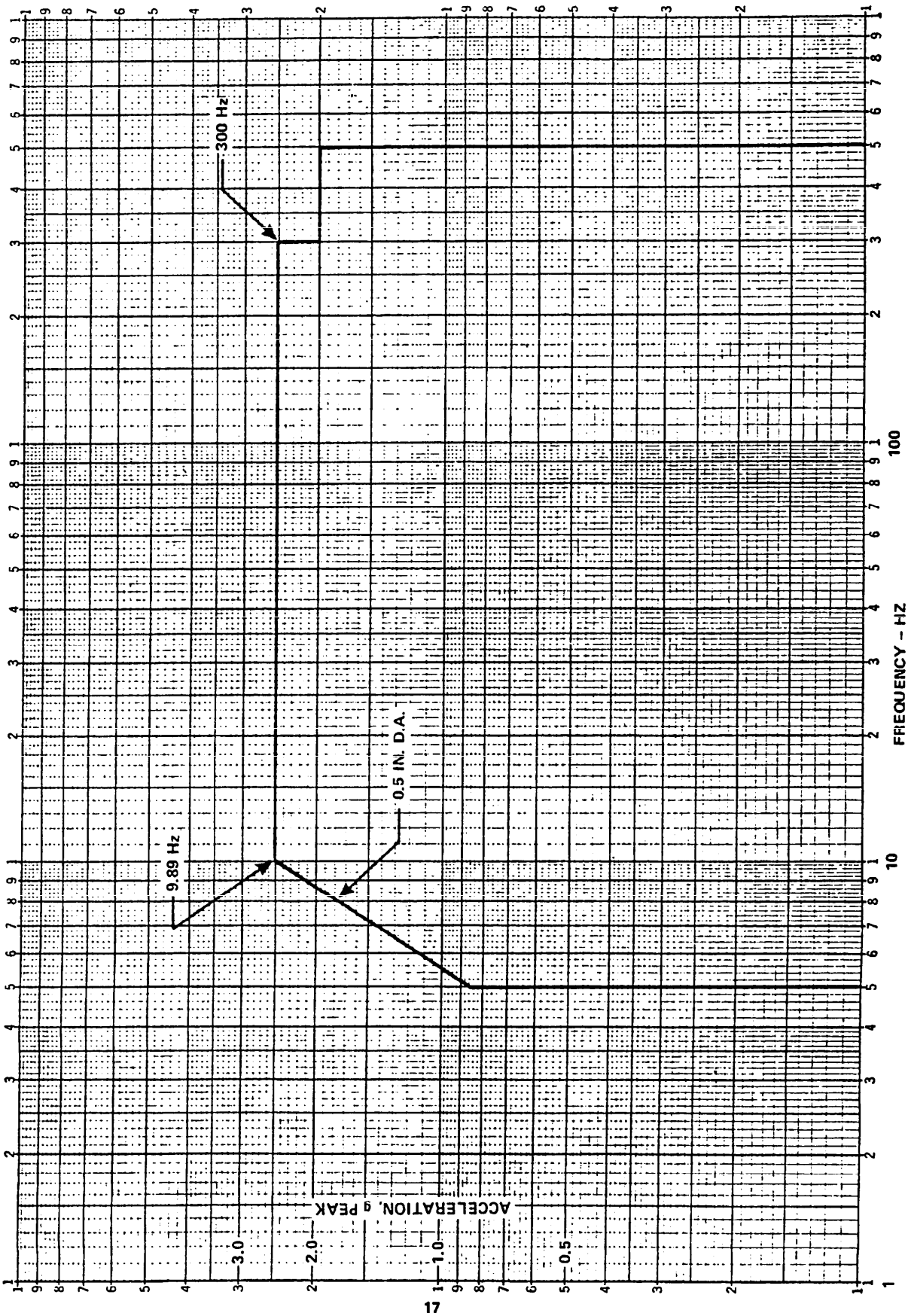


FIGURE 6. Vibration profile.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables IX, X and XI.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

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of supplies in connection with specific procurements

TABLE IX. Performance tests.

Item	Characteristic	Requirement	Test Procedure
101	Lamp test	3.3.1	4.6.2.1
102	$\overline{\text{Select}}$	3.3.2	4.6.2.2
103	$\overline{\text{Overrange}}$	3.3.3	4.6.2.3
104	$\overline{\text{Ripple blanking input (RBI)}}$	3.3.4	4.6.2.4
105	$\overline{\text{Feed}}$	3.3.5	4.6.2.5
106	$\overline{\text{ON}}$	3.3.6	4.6.2.6
107	$\overline{\text{AUTO}}$	3.3.7	4.6.2.7
108	$\overline{\text{TEST switch}}$	3.3.8	4.6.2.8
109	Range select	3.3.9	4.6.2.9
110	$\overline{\text{Manual reset and range light signal}}$	3.3.10	4.6.2.10
111	Turn-on reset	3.3.11	4.6.2.11
112	RANGE	3.3.12	4.6.2.12
113	$\overline{\text{GO}}$, $\overline{\text{Battle range}}$	3.3.13	4.6.2.13
114	Continuity	3.3.14	4.6.2.14

TABLE X. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - visual
302	General specifications	3.1.2	MIL-F-13926 - Visual

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AVU: Inspection level

TABLE XI. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.4.1	4.6.3.1
304	Vibration	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature cycling	3.4.3 and 3.4.4	4.6.3.5

4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table X.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table XI shall be applied separately to each assembly in the sample.

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4.5.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subjected to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table X and only the test of item 307 in table XI shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table IX and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11746960.

4.6 Methods of inspection.

4.6.2 Performance tests.

4.6.1 Test equipment. Unless the assembly is procured apart from the next higher assembly, the test equipment in table XII shall be used where specified to perform the required test.

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TABLE XII. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit, Test Set
2	11750437	Adapter, Switch and Light Control
3	11821457	Fixture Shock, Vertical Horizontal Plane
4	11821458	Shock Adapter, Plate Vib Horz-Vert Plane/Shock-Z
5	11821520	Automatic Test Assembly Central Processor Unit
6	11821647	Vibration/Shock Test Fixture - Electronic Unit, AC, FAT
7	11821861	Test Fixture - Auto Card Test Set, Electronic Unit, FAT

4.6.2.1 Lamp test. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.1.

4.6.2.2 Select. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.2.

4.6.2.3 Overrange. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.3.

4.6.2.4 Ripple blanking input (RBI). Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.4.

4.6.2.5 Feed. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.5.

4.6.2.6 ON. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.6.

4.6.2.7 AUTO. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.7.

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4.6.2.8 TEST switch. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.8.

4.6.2.9 Range select. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.9.

4.6.2.10 Manual reset and range light signal. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.10.

4.6.2.11 Turn-on reset. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.11.

4.6.2.12 RANGE. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.12.

4.6.2.13 GO, Battle range. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.13.

4.6.2.14 Continuity. Use items 1, 2 and 5 of table XII or suitable equipment to determine compliance with 3.3.14.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use items 3, 4 and 6 of table XII or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 ±4 g with a time duration of 18 ±3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table IX.

4.6.3.1.2 High intensity. Use items 3, 4 and 6 of table XII or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 120 ± 12 g with a time duration of 1.7 ±0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table IX.

4.6.3.2 Vibration. Vibration A as specified in 4.6.3.2.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.3.2.2 shall be conducted only for quality conformance inspection.

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4.6.3.2.1 Vibration A. Use items 4 and 6 of table XII or mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2 procedure VIII, except that the test level shall be the vibration curve shown in figure 1 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table IX.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Use item 7 of table XII or suitable equipment and subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 125°F. The performance tests of table IX shall be applied.

4.6.3.4 Low temperature. Use item 7 of table XII or suitable equipment and subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table IX shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 7. At the conclusion of this test, subject the assembly to the performance tests of table IX.

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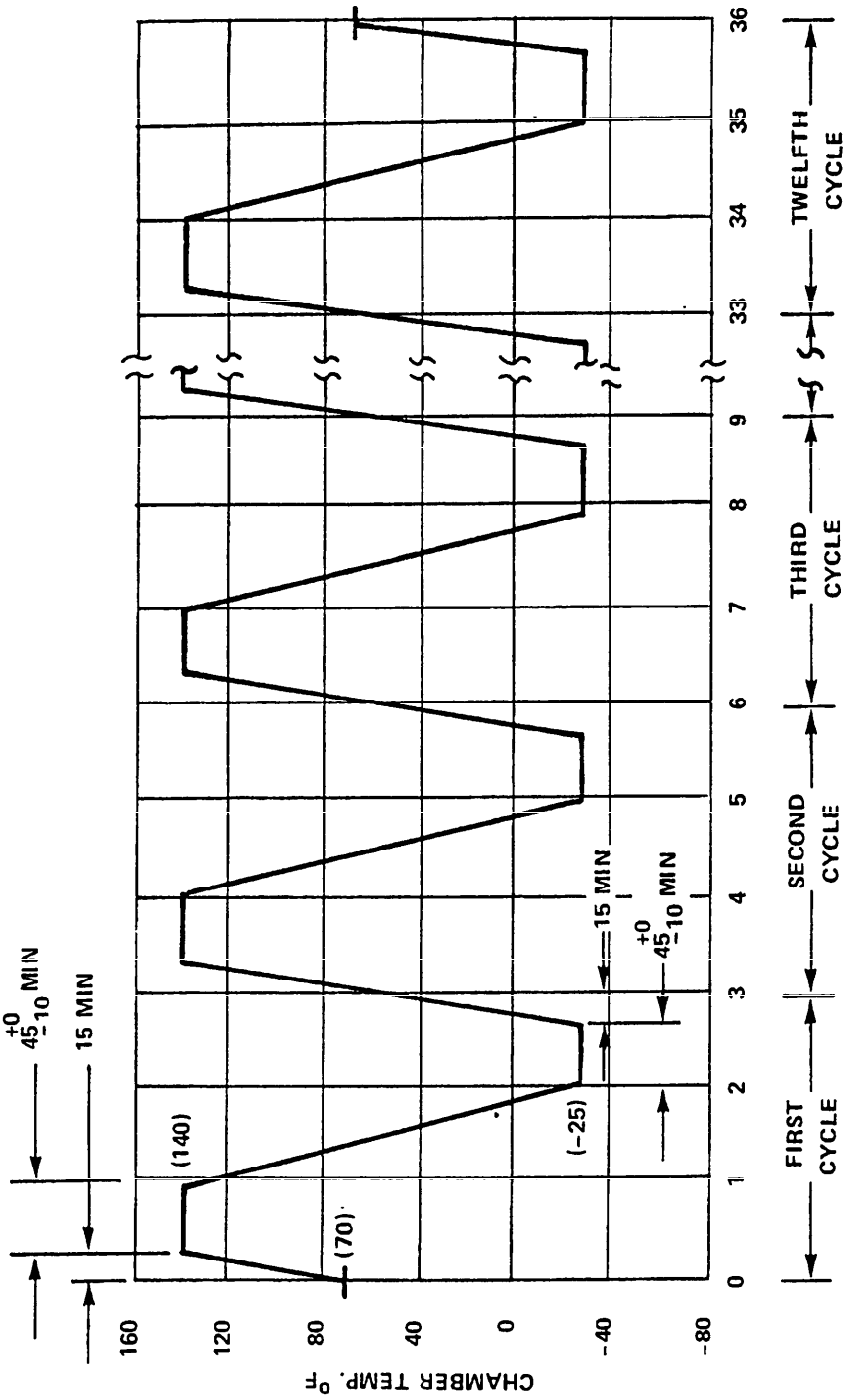


FIGURE 7. Temperature cycling.

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5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11746960. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The Switch and Light Control Circuit Card Assembly is a functional assembly of the Electronic Unit intended for use in the Laser Fire Control Rangefinder (AN/VVG-2). The assembly provides logic for:

a. Range selections
b. Overrange detection
c. Lamp control
d. Computer

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Special tests and examination as required, the type of inspection required and the activity responsible for testing and its facility.

c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. First article inspection requirements.

6.3 Previous revisions of MIL-C-48601A was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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